Snowflake(Exercise)

Friday, August 01, 2025 11:31 AM

Conceptual Questions

1. What are the three layers of Snowflake's architecture, and what does each handle?
2. Explain the difference between Permanent, Transient, and Temporary tables.
3. What is the purpose of a Virtual Warehouse in Snowflake?
4. How does Snowflake handle semi-structured data like JSON?
5. Compare Snowpipe with the COPY INTO command.

Hands-On Exercises

1. Create a database `retail_db` and a schema `q3_sales`.
2. Create a Virtual Warehouse named `retail_wh` with size SMALL, auto-suspend = 300 seconds, auto-resume = TRUE.
3. Create a permanent table `products` with columns: product_id, product_name, price.
4. Insert 5 rows of sample data into `products`.
5. Query all rows where price > 100.

Intermediate Level

Conceptual Questions

1. What is the difference between Time Travel and Zero-Copy Cloning in Snowflake?
2. How do Internal Stages differ from External Stages?
3. What role does RBAC play in Snowflake, and why is it important?
4. How do Streams help implement Change Data Capture (CDC)?
5. Explain how Snowflake achieves separation of compute and storage.

Hands-On Exercises

1. Create a file format `my_csv_format` for CSV files (skip 1 header row).
2. Create an internal stage `my_stage` and upload the sample file `sales.csv`.
3. Load the staged file into a Snowflake table `sales_fact`.
4. Use the VARIANT data type to load and query a JSON file `customers.json`. Extract customer names from the nested JSON.
5. Create a Stream on `sales_fact` and query new rows after inserting additional data.
6. Create a Task that runs every 5 minutes to MERGE new rows from the stream into a summary table.

Advanced Level

Conceptual Questions

1. In which scenarios would you use Snowpipe Streaming instead of classic Snowpipe?
2. Describe how to secure shared data using Secure Views and Row Access Policies.
3. What are some best practices for designing role hierarchies in Snowflake?
4. How does Snowflake handle semi-structured data flattening with LATERAL FLATTEN? Provide an example.
5. Explain the billing model in Snowflake — what are the cost drivers for compute vs. storage?

Hands-On Exercises

1. Create a table `orders_raw` and a Snowpipe `orders_pipe` that auto-ingests from an external S3 stage.
2. Query the Snowpipe metadata to verify successful ingestions.
3. Use Time Travel to recover a table dropped accidentally.
4. Clone the `sales_fact` table as `sales_clone` and demonstrate that the clone initially uses no extra storage.
5. Create a Masking Policy to hide email addresses unless the user has the HR_ROLE. Test with two different roles.
6. Create a Row Access Policy that allows a user to see only rows for their assigned region.



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